HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** REST APIs for enterprise systems, handling approximately **1,500 requests per minute** during peak hours.
- Designed and implemented database schemas and optimized queries across **PostgreSQL** and **MySQL** for core business applications, supporting roughly **200 daily operational reports**.
- Authored **Python scripts** for automating ETL processes and data validation, reducing manual data transformation time by an estimated **30%** for the financial operations team.
- Supported the manufacturing line by performing root cause analysis on electronic medical devices, identifying component-level failures and contributing to **quality improvements**.
- Collaborated with a **cross-functional team** including quality and manufacturing engineers to resolve production issues and enhance test equipment yields.
- Performed tests, analyzed data, and authored reports to validate electronic manufacturing systems against industry standards, utilizing **standard applications** and in-house tracking systems.
- Containerized several internal Java services using **Docker** and deployed them to **AWS EC2** instances, improving deployment consistency across development and staging environments.

SKILLS

Programming Languages – Java, Python, SQL, C++, C

Frameworks & Libraries – Spring Boot, Scikit-learn, NumPy, pandas

Databases – PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Tools & Technologies – Jira, Confluence, Splunk, Dynatrace, Linux, Power BI, Tableau, LabVIEW

PROJECTS

Automated Test Data Generation for Medical Device Software

- Developed a **Python** application to generate synthetic test data for implantable medical device software, simulating various operational parameters and failure modes.
- Integrated the data generation tool with CI/CD pipelines using **Jenkins**, enabling automated regression testing for new software releases.
- Created comprehensive documentation and provided training sessions for the test engineering team on utilizing the generated data for validation activities.

Production System Monitoring Dashboard

- Built a **Java Spring Boot** microservice to aggregate real-time monitoring data from various production systems, including error rates and system health metrics.
- Designed and implemented RESTful APIs to expose this aggregated data, which was then visualized on internal dashboards for operations teams.
- Wrote unit tests using **JUnit**, achieving over **85% code coverage** for the core data aggregation and API modules.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)